

LOW ACRYLONITRILE NBR: PATENT LANDSCAPE ON NITRILE RUBBER METATHESIS AND COMPOSITION-CONTROLLED FEEDSTOCKS

1. ABSTRACT

This report reviews two primary patents relevant to low-acrylonitrile nitrile butadiene rubber (NBR) in the provided manifest, with emphasis on how low-ACN NBR feedstocks are handled in downstream processing rather than direct low-ACN copolymer synthesis. Both patents define NBR broadly as a copolymer or terpolymer of an α,β -unsaturated nitrile and a conjugated diene, optionally with additional copolymerizable monomers, and both focus on metathesis-based molecular-weight reduction of nitrile rubber prior to hydrogenation. The technical target in this landscape is not a new monomer recipe for low acrylonitrile incorporation, but rather the preparation of lower-molecular-weight NBR intermediates that can be hydrogenated to HNBR while preserving processability and avoiding excessive viscosity rise.

US7875683B2 discloses metathetic degradation of nitrile rubber using a metal-carbene metathesis catalyst combined with a transition-metal additive of general formula $M(OZ)_m$, aiming to improve degradation efficiency and lower molecular weight. The patent emphasizes the challenge of producing sufficiently low-Mn NBR feedstocks for hydrogenation and notes practical limits around Mooney viscosity and work-up. EP2028194B1 discloses a related metathesis process using specific ruthenium/osmium complex catalysts, with comparative examples showing that a Grubbs III catalyst outperforms a Grubbs II catalyst at the same ruthenium loading and 1-hexene co-olefin level, giving lower Mw and narrower PDI in shorter times.

Across the two patents, the dominant process trend is olefin metathesis of NBR in the presence of a co-olefin, followed by hydrogenation, with catalyst selection being the main lever for improving activity, lowering molecular weight, and narrowing dispersity. No patent in the manifest directly discloses a low-acrylonitrile copolymerization recipe; instead, they provide enabling degradation/process technology for NBR grades that may include low-ACN feedstocks.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US7875683B2
- Patent Title: Process for the metathetic degradation of nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses molecular-weight reduction of nitrile rubber by metathesis using a metal-carbene complex catalyst based on a transition metal of group 6 or 8 together with a compound of general formula $M(OZ)_m$ as an additive. The process is presented as a way to improve metathetic degradation efficiency for NBR prior to hydrogenation, thereby enabling lower-molecular-weight feedstocks. - Catalyst metal: transition group 6 or 8 — Claim 1 - Additive metal: transition group 4, 5 or 6 — Claim 1 - Additive formula: $M(OZ)_m$ — Claim 1 - m value: 4, 5 or 6 — Claim 1 - Z radical carbon count: 1-32 carbon atoms — Claim 1 - Z heteroatoms: 1-15 heteroatoms — Claim 1 - NBR feed Mooney viscosity: about 30 Mooney units (ML 1+4 at 100°C) — Background - Hydrogenated nitrile rubber Mooney viscosity: in the order of 55 Mooney units (ML 1+4 at 100°C) — Background - Target degraded NBR Mooney viscosity: less than 30 Mooney units (ML 1+4 at 100°C) — Background - Target degraded NBR number average molecular weight: $M_n < 70\,000$ g/mol — Background - HNBR degree of hydrogenation: 50 to 100% — Background - HNBR residual double bond content: 1 to 18% — Background - HNBR Mooney viscosity: 55 to 120 — Background - HNBR number average molecular weight: about 200 000 to 700 000 — Background

Relevant Experimental Evidence - Example highlights: No numbered examples were detected in the provided extraction; the patent text is primarily claim- and background-driven. - Example highlights: The disclosure establishes that metathesis is used to reduce NBR molecular weight before hydrogenation. - Example highlights: The patent frames the additive/catalyst combination as improving degradation efficiency relative to prior metathesis systems.

Technical Relevance This patent is relevant because low-acrylonitrile NBR grades are still NBR feedstocks, and the disclosed metathesis degradation route addresses the key processing problem of lowering molecular weight before hydrogenation. It provides a catalyst/additive strategy for making NBR intermediates more processable, which is important when composition-controlled low-ACN feedstocks must be converted into lower-viscosity derivatives.

Patent 2

- Patent Number: EP2028194B1

- Patent Title: Metathesis of nitrile rubbers in the presence of transition metal complex ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses metathetic degradation of nitrile rubber in the presence of specific ruthenium/osmium complex catalysts, optionally with 1-hexene as co-olefin, to reduce molecular weight before subsequent hydrogenation. The examples compare Grubbs III and Grubbs II catalysts and show that the newer catalyst gives faster degradation and narrower molecular-weight distribution at the same ruthenium loading. - Ruthenium loading: 23 ppm — Example 1 - 1-Hexene: 2.0 phr — Example 1 - NBR amount: 80 g — Example 1 - Catalyst amount: 0.02 g — Example 1 - Reaction temperature: 40 °C — Example 1 - Reaction time: 0 min, 15 min, 60 min — Example 1 - Mw: 240 kg/mol — Example 1 - Mw: 150 kg/mol — Example 1 - Mw: 144 kg/mol — Example 1 - Mn: 100 kg/mol — Example 1 - Mn: 71 kg/mol — Example 1 - Mn: 72 kg/mol — Example 1 - PDI: 2.4 — Example 1 - PDI: 2.1 — Example 1 - PDI: 2.0 — Example 1 - Ruthenium loading: 57 ppm — Example 2 - 1-Hexene: 2.0 phr — Example 2 - NBR amount: 200 g — Example 2 - Catalyst amount: 0.05 g — Example 2 - Reaction temperature: 40 °C — Example 2 - Reaction time: 0 min, 15 min, 90 min — Example 2 - Mw: 240 kg/mol — Example 2 - Mw: 39 kg/mol — Example 2 - Mw: 36 kg/mol — Example 2 - Mn: 100 kg/mol — Example 2 - Mn: 21 kg/mol — Example 2 - Mn: 20 kg/mol — Example 2 - PDI: 2.4 — Example 2 - PDI: 1.8 — Example 2 - PDI: 1.8 — Example 2

Relevant Experimental Evidence - Example 1: Grubbs III catalyst at 23 ppm Ru with 2.0 phr 1-hexene reduced Mw from 240 to 144 kg/mol and narrowed PDI to 2.0 within 60 min. - Example 1: Comparative Grubbs II catalyst at the same Ru and 1-hexene level gave slower degradation and higher final Mw (168 kg/mol at 90 min). - Example 2: Grubbs III catalyst at 57 ppm Ru and 2.0 phr 1-hexene reduced Mw from 240 to 36 kg/mol and Mn to 20 kg/mol within 90 min. - Example 2: Comparative Grubbs II catalyst at the same Ru and 1-hexene level gave much higher Mw (65 kg/mol at 90 min) and broader PDI than Grubbs III. - Example 2: The degraded nitrile rubbers obtained with Grubbs III were gel-free.

Technical Relevance This patent is relevant because it provides a more active metathesis catalyst system for lowering NBR molecular weight, which is directly useful for preparing processable NBR feedstocks that may be low in acrylonitrile. The comparative data identify catalyst choice, co-olefin use, and temperature as key levers for achieving low-Mw NBR efficiently.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- Both patents address NBR degradation by olefin metathesis rather than direct low-ACN copolymerization; neither discloses a monomer-feed recipe for lowering acrylonitrile incorporation.
- US7875683B2 uses a metathesis catalyst plus a transition-metal additive $M(OZ)_m$, while EP2028194B1 uses defined ruthenium/osmium complex catalysts without the additive strategy.
- EP2028194B1 provides explicit example data with 1-hexene co-olefin, 40 °C reaction temperature, and catalyst loadings of 23 ppm and 57 ppm Ru; US7875683B2 provides no numbered examples or comparable operating data in the extracted evidence.
- EP2028194B1 demonstrates that Grubbs III outperforms Grubbs II at identical Ru loading, giving lower Mw and narrower PDI; US7875683B2 instead claims improved degradation through catalyst/additive synergy but without example-level comparison in the extraction.
- Both patents frame the practical target as lowering NBR Mooney viscosity and molecular weight before hydrogenation, with the background noting a feedstock Mooney viscosity around 30 and a degraded target below 30 ML 1+4 at 100 °C in US7875683B2.
- Co-olefin use is explicit in EP2028194B1 (1-hexene, 2.0 phr), while US7875683B2 mentions low molecular weight 1-olefins generally but does not provide a specific example in the extracted text.

4. CONCLUSION

The patent evidence indicates that the most relevant synthesis route for low-acrylonitrile NBR in this manifest is not direct low-ACN monomer copolymerization, but metathesis-based molecular-weight reduction of NBR feedstocks prior to hydrogenation. The strongest operational evidence comes from EP2028194B1, where a Grubbs III catalyst with 1-hexene at 40 °C and 23-57 ppm Ru gives substantially lower Mw and narrower PDI than Grubbs II. US7875683B2 supports the same general process direction and adds a transition-metal additive strategy to improve metathesis efficiency. Based on the evidence, the recommended process window is ruthenium-based metathesis catalysis, optional low-olefin co-olefin addition, and careful control of catalyst activity to drive NBR to below 30 Mooney units or below 70,000 g/mol Mn before hydrogenation.

5. REFERENCES

- US7875683B2 | Process for the metathetic degradation of nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7875683B2/en>
- EP2028194B1 | Metathesis of nitrile rubbers in the presence of transition metal complex ... | Not disclosed | EP | 2010 | <https://patents.google.com/patent/EP2028194B1/en>